



Supply Chain Management

Fundamentals, Frameworks, and Real-World Applications — Software & Hardware

Today's Topic

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What Is a Supply Chain?

Definition, objectives, and why it matters

02

Decision Phases & Process Views

Strategic, planning, and operational decisions

03

Macro Processes in a Firm

CRM, ISCM, and SRM explained

04

SCM Fundamentals

Software and hardware supply chain specifics

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Real-World Examples & Applications

Applying principles to product and process design



PART 1

What Is a Supply Chain?

A supply chain encompasses **all parties involved** — directly or indirectly — in fulfilling a customer request. It includes manufacturers, suppliers, transporters, warehouses, retailers, and customers themselves.

The Objective of a Supply Chain

Core Goal

Maximize overall **supply chain surplus** — the difference between the value a customer receives from a product and the total cost incurred across the entire chain.

$$\text{SC Surplus} = \text{Customer Value} - \text{Total SC Cost}$$

What This Means in Practice

- Every decision should **increase profitability** across the chain — not just for one firm
- Cost reduction alone is not enough — **value creation** matters equally
- Collaboration between stages leads to **greater surplus** than competition
- Metrics: total revenue minus costs of inventory, transport, operations, and information

Decision Phases in a Supply Chain

Supply chain decisions occur at three time horizons, each with different levels of commitment and reversibility.



Supply Chain Strategy

Long-term. Structure of the chain: how many warehouses, which suppliers, outsource or in-house? Decisions are expensive to reverse.



Supply Chain Planning

Medium-term (quarterly/annual). Which markets to serve, inventory policies, promotions, and subcontracting. Operates within strategic constraints.



Supply Chain Operations

Short-term (daily/weekly). Fulfilling individual orders, routing shipments, and managing real-time inventory. Goal: minimize uncertainty.

Process Views of a Supply Chain

Two Complementary Perspectives

Supply chains can be analyzed through two lenses — both are essential for understanding how value flows from raw material to end customer.

Cycle View

Processes divided into cycles between successive stages: **Customer Order → Replenishment → Manufacturing → Procurement**. Clearly defines roles and responsibilities at each interface.

Push/Pull View

Pull processes are initiated by a customer order (reactive). **Push** processes are initiated in anticipation of demand (speculative). The boundary between them is the **push-pull boundary**.

Example: Dell vs. Traditional PC Retailer

Dell (Pull-Heavy)

Customer places order → Dell assembles to order. Almost entirely **pull**. Minimal finished-goods inventory. Push only at component level.

Traditional Retailer (Push-Heavy)

Manufacturer forecasts demand → builds inventory → ships to stores. Mostly **push**. Higher inventory risk but faster availability.



PART 2

Supply Chain Macro Processes in a Firm

All supply chain activities within a company can be grouped into three macro-level processes that span functional boundaries and connect external partners to internal operations.

The Three Macro Processes



CRM — Customer Relationship Management

All processes at the **customer-facing interface**. Includes marketing, sales, order management, and call centers. Goal: generate and fulfill customer demand effectively.



ISCM — Internal Supply Chain Management

All processes **internal to the firm**: strategic planning, demand planning, supply planning, fulfillment, and field service. Balances supply and demand inside the organization.



SRM — Supplier Relationship Management

All processes at the **supplier-facing interface**. Includes supplier evaluation, selection, negotiation, and collaboration. Goal: obtain the right inputs at the right time and cost.

- ❑ Integration across CRM, ISCM, and SRM is the key to achieving competitive supply chain performance. Silos between these processes are a primary source of inefficiency.

Fundamentals of Supply Chain Management



Flow of Goods, Information & Funds

SCM coordinates the **three flows**: physical product movement, information sharing (demand signals, forecasts), and financial transactions. All three must be synchronized.



Integration & Coordination

Information sharing across partners reduces the **bullwhip effect** — where small demand fluctuations amplify into huge inventory swings upstream. Coordination is the cure.



Efficiency vs. Responsiveness

The central tradeoff. **Efficient chains** minimize cost (commodity products, stable demand). **Responsive chains** react quickly to uncertainty (fashion, tech, custom orders).



Performance Measurement

Key metrics include **fill rate, order cycle time, inventory turns, total cost, and cash-to-cash cycle time**. What gets measured gets managed.



PART 4

Software Supply Chain Management

Software has its own supply chain — from **open-source components** and **third-party APIs** to deployment pipelines and end-user delivery. Managing it rigorously is critical for quality and security.

Software Supply Chain: Key Concepts

What Makes Software SCM Unique

Unlike physical goods, software supply chains involve **digital dependencies** — libraries, packages, and services — that are versioned, patched, and distributed globally. The "product" can be replicated at near-zero cost but carries compounding risk at each dependency layer.

1 Dependency Management

Tracking and vetting open-source libraries, SDKs, and APIs used in a codebase (e.g., npm, Maven, PyPI packages). A vulnerability in one dependency can cascade across thousands of products.

2 CI/CD Pipelines as Production Lines

Continuous Integration/Continuous Delivery pipelines are the software equivalent of a manufacturing line. Code moves through build, test, staging, and release stages — each is a supply chain node.

3 SBOM — Software Bill of Materials

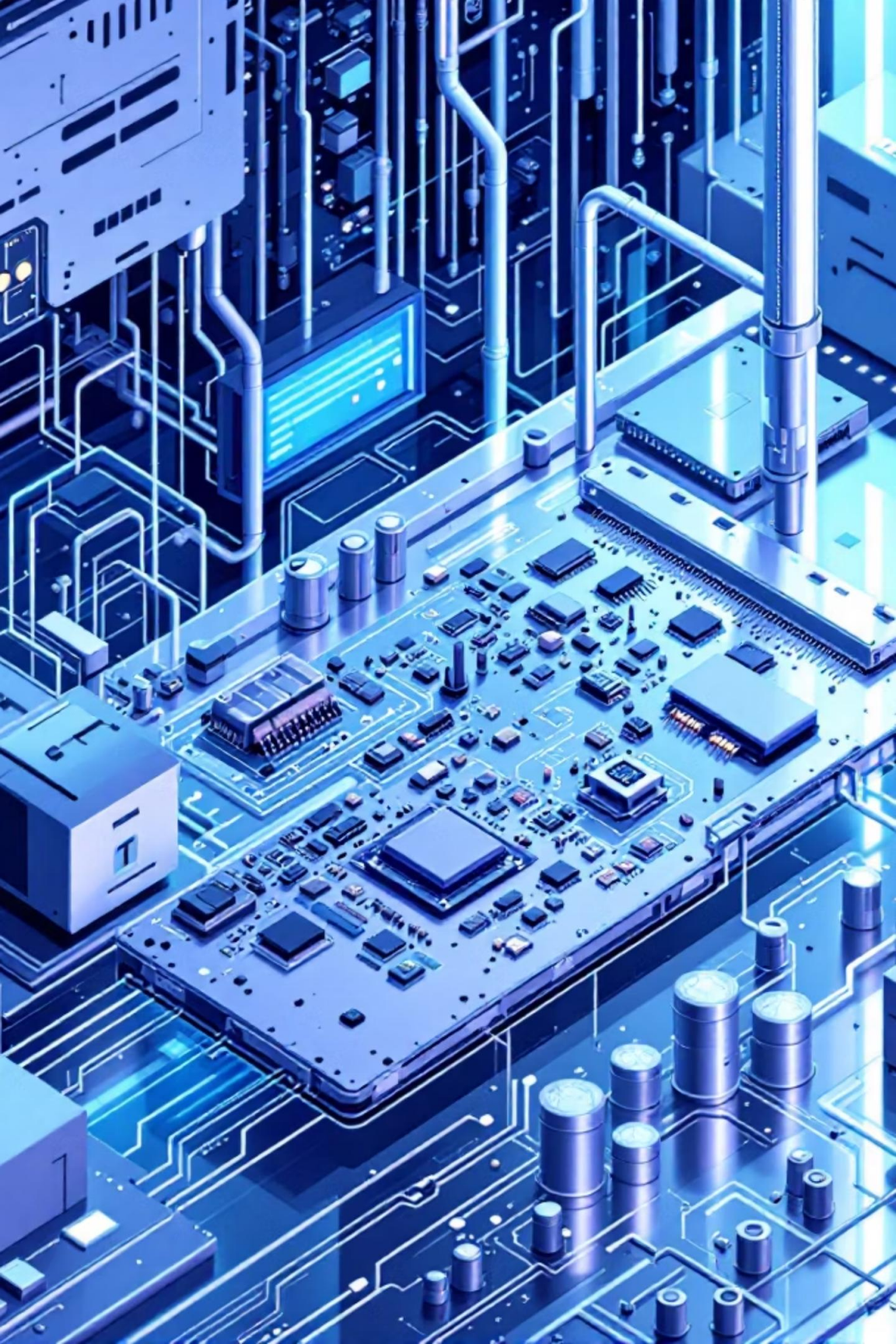
A formal inventory of all components in a software product — analogous to a manufacturing BOM. Increasingly required for government and enterprise procurement.

Real-World Risk Example

SolarWinds Attack (2020): Hackers compromised the software build process of SolarWinds, inserting malicious code into a routine software update. Over 18,000 organizations — including U.S. federal agencies — unknowingly installed the backdoor.

This is a **supply chain attack**: targeting an upstream supplier to reach downstream customers. It exposed how critical software SCM security has become.

❏ Key lesson: your software supply chain is only as secure as its weakest upstream component.

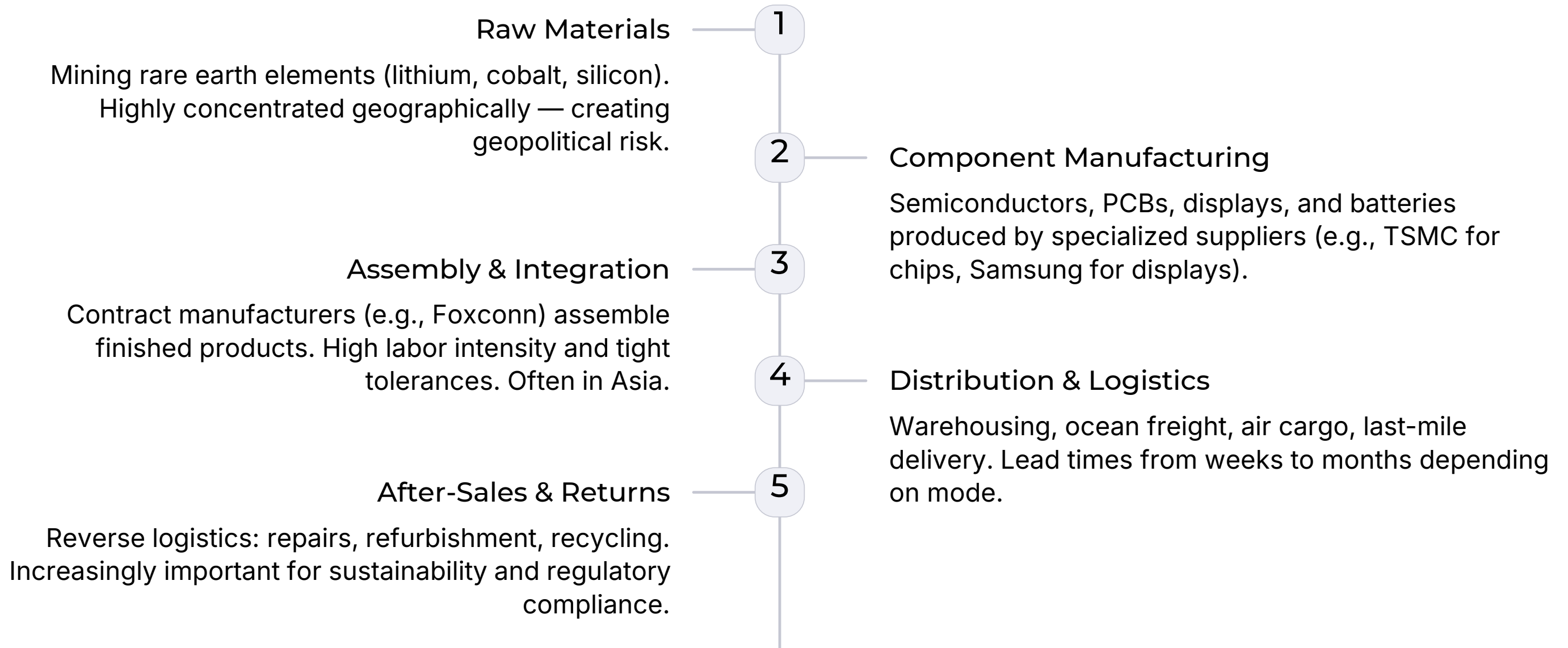


PART 5

Hardware Supply Chain Management

Hardware supply chains are among the most **globally complex and capital-intensive** systems in modern industry — spanning raw material extraction, component fabrication, assembly, distribution, and after-sales service across dozens of countries.

Hardware Supply Chain: Key Concepts



Hardware SCM requires **long-horizon planning** — chip lead times can exceed 52 weeks — making demand forecasting and safety stock strategy critical decisions.

Real-World Supply Chain Examples



Apple iPhone

Over **200 suppliers across 43 countries**. Chips from TSMC (Taiwan), displays from Samsung (Korea), assembly by Foxconn (China). Apple maintains tight supplier scorecards — a textbook SRM practice.



Amazon Fulfillment

Amazon's supply chain is its competitive moat. **Predictive inventory placement**, robotics-driven warehouses, and same-day delivery networks exemplify a highly responsive, data-driven push-pull system.



Toyota Production System

The gold standard of **lean supply chain design**. Just-in-Time delivery, kanban signals, and deep supplier partnerships eliminate waste while maintaining flexibility — a model studied globally.

Applying SCM Principles to Software & Hardware Development

Software Development Process Design

- **Treat CI/CD as your production line** — apply lean principles: reduce waste (bugs, rework), increase flow, measure throughput
- **Manage your SBOM** like a procurement list — vet, approve, and monitor every dependency
- **Pull-based deployment**: release when customer demand (user stories) pulls features through the pipeline
- **Inventory analogy**: unshipped code is WIP inventory — minimize it by deploying frequently

Hardware Development Process Design

- **BOM planning early**: engage component suppliers at design stage to avoid sourcing surprises at launch
- **Design for Supply Chain**: choose standard, widely available components; avoid single-source dependencies
- **Safety stock strategy**: buffer critical long-lead-time parts; use risk pooling for common components
- **Supplier qualification**: apply SRM principles — audit quality, capacity, financial stability, and geopolitical exposure

- ❏ The same SCM principles — integration, flow, tradeoffs, and measurement — apply whether your "product" ships in a box or downloads from the cloud.

The Bullwhip Effect: A Visual Lesson

What It Is

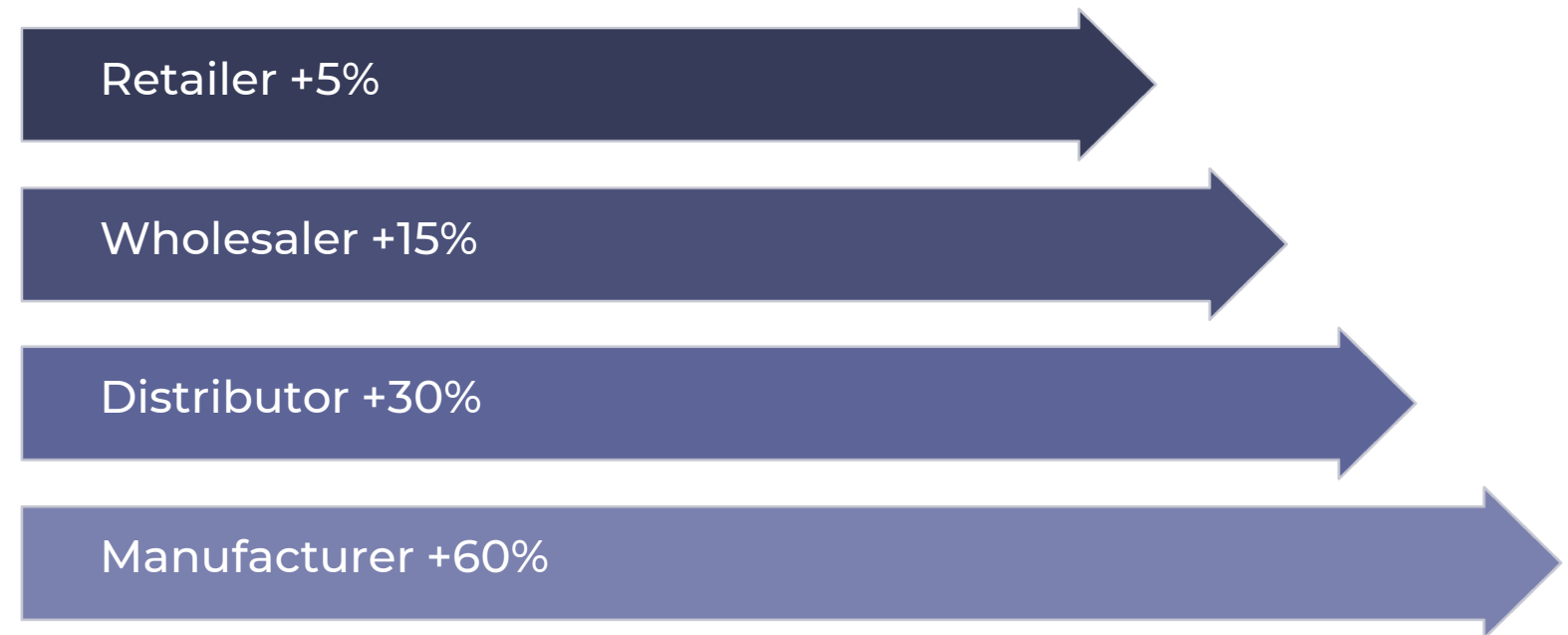
A small change in end-customer demand causes progressively **larger swings in orders** as you move upstream — from retailer to wholesaler to manufacturer to supplier. The result: excess inventory, stockouts, and wasted capacity.

Root Causes

- Demand forecast errors at each stage
- Order batching (ordering in large lots)
- Price fluctuations and promotions
- Rationing and shortage gaming

The Fix

Information sharing (share POS data upstream), **VMI** (vendor-managed inventory), and **collaborative forecasting (CPFR)** are proven remedies.



Key Takeaways

Supply Chain = Network of Value

Every party from raw material to end customer is part of the chain.
Maximizing **total surplus** — not individual profit — is the objective.

Decisions Happen at Three Horizons

Strategy, Planning, and Operations each require different tools, timelines, and trade-offs.
Misalignment across horizons is a common failure point.

Push vs. Pull Defines Your Risk Profile

More push = more inventory risk.
More pull = more responsiveness.
The right balance depends on your product's demand uncertainty and customer expectations.

SCM Applies to Software Too

Dependencies, pipelines, and deployment cycles are supply chain problems. Managing them with SCM rigor improves quality, speed, and security.

Integration Is the Competitive Advantage

CRM + ISCM + SRM working together — supported by real-time data — is what separates world-class supply chains from average ones.

Discussion Questions

1

Push or Pull?

A startup is launching a new smartphone. Should they build to stock (push) or build to order (pull)? What factors should drive that decision?

2

Software SBOM

Your company uses 47 open-source libraries in its product. One has a critical security vulnerability. How does SCM thinking help you respond systematically?

3

Bullwhip in Tech

During COVID-19, the semiconductor industry experienced a severe chip shortage. Which bullwhip causes do you think contributed most? What would have helped?

4

Macro Processes

At a company like Netflix, how would you define their CRM, ISCM, and SRM? Is the framework different for a digital-native firm versus a manufacturer?

